

2.2 System Architecture Overview

Introduction

The Online Tutor Marketplace system is designed using a multi-layered web-based architecture to ensure efficient communication between users, system components, and data resources. The architecture supports functionalities such as user authentication, tutor searching, booking management, messaging, AI-based recommendations, and administrative monitoring.

The proposed architecture improves scalability, maintainability, system performance, and data management by separating the frontend, backend, database, and AI processing components into distinct layers.

System Architecture Components

The Online Tutor Marketplace system consists of four major architectural components:

- User Layer
- Frontend Layer
- Backend Layer
- Database Layer

The system also integrates the OpenAI API to provide intelligent tutor recommendations.

User Layer

The user layer represents external users who interact with the system including students, tutors, and administrators.

Students can search tutors, make bookings, communicate through the messaging system, and receive AI-based tutor recommendations.

Tutors can manage profiles, respond to booking requests, and update availability.

Administrators are responsible for monitoring platform activities and managing user accounts.

Frontend Layer

The frontend layer represents the graphical user interface where users interact with the system. The interface includes login pages, dashboards, tutor search pages, booking pages, messaging systems, and AI recommendation interfaces.

The frontend is designed to provide a clean, responsive, and user-friendly experience across different devices and browsers.

Technologies such as HTML, CSS, JavaScript, and Figma-based UI design principles are used to support the frontend implementation.

Backend Layer

The backend layer handles application logic and system processing. It performs authentication, tutor search processing, booking operations, messaging functionality, AI recommendation processing, and administrative operations.

The backend validates user requests and communicates with the database to retrieve or update information.

The backend also communicates with the OpenAI API to generate intelligent tutor recommendations.

Database Layer

The database layer stores and manages system information including user accounts, tutor profiles, bookings, reviews, and messages.

The database ensures secure storage, efficient retrieval, consistency, and relationship management using primary and foreign keys.

AI Integration Component

The system integrates an AI-based recommendation feature using the OpenAI API. The AI component recommends tutors based on subject requirements, student learning level, tutor ratings, and availability.

System Interaction Flow

1. Users interact with the frontend interface.
2. User requests are sent to the backend layer.
3. The backend validates and processes requests.
4. The backend communicates with the database layer.
5. AI requests are processed using the OpenAI API.
6. Processed results are returned to the frontend and displayed to users.

This structured architecture ensures efficient communication, reliable system performance, and secure data processing across all platform components.

System Architecture Diagram

Insert the final architecture diagram below this section. The diagram should include the User Layer, Frontend Layer, Backend Layer, Database Layer, and OpenAI API integration with clear interaction flow between components.

Explanation of Architecture Diagram

The system architecture diagram illustrates the interaction between the frontend, backend, database, and AI components of the Online Tutor Marketplace system. User requests are initiated through the frontend interface and processed by the backend modules. The backend communicates with the database layer to store and retrieve information required by the system.

The AI processing module interacts with the OpenAI API to generate intelligent tutor recommendations based on user learning requirements and preferences. The processed results are then returned to the frontend and displayed to users.

The proposed system architecture supports efficient communication between system components while ensuring scalability, maintainability, secure data management, and effective integration of AI-based functionalities within the Online Tutor Marketplace platform.